

Model Fairness, Bias & Explainability Report

Student Name	Dhanavath Navadeep	Internship	CodeAlpha
Project Title	Model Fairness, Bias & Explainability	Dataset Used	Adult Income Dataset
GitHub Repo	github.com/DhanavathNavadeep/Task_3_Model_Fairness_Bias_Explainability		

Project Objective

The objective of this project was to develop a machine learning classification model, systematically evaluate its fairness across sensitive groups, and generate actionable explainability outputs. This technical approach ensures increased transparency, auditability, and clear interpretability of model predictions.

Tools and Technologies

Python

Pandas

Scikit-learn

Fairlearn

LIME

Matplotlib

Joblib

VS Code

Git & GitHub

Project Workflow

- **Data Ingestion & Preprocessing:** Loaded and preprocessed the raw Adult Income Dataset.
- **Feature Engineering:** Transformed text dimensions by encoding categorical features into algorithmic representations.
- **Data Splitting:** Divided the preprocessed dataset into structured training and testing matrices.
- **Model Architecture:** Initialized and trained a robust Random Forest Classifier.
- **Performance Evaluation:** Checked model performance criteria and evaluated general predictive accuracy.
- **Fairness Auditing:** Conducted programmatic demographic parity and disparate impact evaluations using Fairlearn.
- **Explainability Assessment:** Generated model transparency diagnostics including Global Feature Importance and Local Interpretable Model-agnostic Explanations (LIME).
- **Serialization & Deployment:** Saved all persistent artifacts and synchronized the workspace codebase via Git to GitHub.

Results & Findings

- **Model Optimization:** Random Forest Classifier optimized and trained successfully.
- **Predictive Accuracy:** Achieved a solid validation baseline accuracy of **75%**.
- **Fairness Metrics:** Disparate impact reports and group-level performance matrices successfully generated.
- **Global Explanations:** Feature Importance visualization successfully mapped out top criteria driving predictions.
- **Local Explanations:** Highly descriptive individual LIME explanation visualizations compiled.
- **Artifact Serialization:** Trained predictive intelligence serialized securely as `model.pkl`.

Project Output Files

File / Artifact	Functional Description
<code>model.pkl</code>	Trained Random Forest Classifier binary saved using Joblib.
<code>fairness_metrics.csv</code>	Tabular tracking of metric deviations across demographics.
<code>bias_report.txt</code>	Detailed analytical summary regarding demographic parity vectors.
<code>feature_importance.png</code>	Graph visualizing global weights assigned to features.
<code>lime_explanation.png</code>	Local sample prediction explanation chart mapping input factors.
<code>shap_force_plot.html</code>	Interactive HTML dashboard evaluating interactive SHAP value forces.

Challenges Faced & Solutions Implemented

1. Resolution of Execution Environment Path Inconsistencies

Challenge: Encountered incorrect relative file paths while executing scripts that require loading the trained model binary.

Solution: Updated the training workflow script structure to establish absolute and runtime-resilient saving rules, ensuring proper serialization paths.

2. Version Control & Structure Alignment

Challenge: Faced workspace architecture design problems and Git repository structure conflicts.

Solution: Re-architected directory components, resolved staging configurations, and cleanly re-pushed matching code structures.

3. Pipeline Pipeline Breaking Due to Missing Artifacts

Challenge: Missing model binary instances during downstream runtime execution of fairness and explainability pipelines.

Solution: Synchronized step execution order, fixed missing script triggers, and successfully regenerated all evaluation outputs.

Conclusion

This project successfully demonstrates the comprehensive implementation of an operational machine learning model combined with rigorous fairness evaluations and modern transparency techniques. The compiled reports, metrics, and explanatory visualizations substantially improve accountability pipelines, making it easier to audit models and mitigate systemic bias vectors in live operational frameworks.